Cell Structure

Exam code: 0610 & 0970

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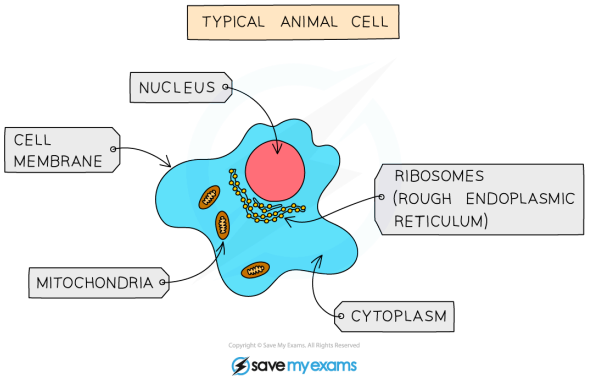
Help me understand

Animal & plant cells

Animal cell structure

- **The main features of animal cells:**
  - They contain a **nucleus** with a **distinct membrane**
  - **Cells do not** have **cellulose cell walls**
  - Their cells **do not** contain **chloroplasts** (so they **are unable** to carry out **photosynthesis**)
  - They contain carbohydrates stored as **glycogen**

Animal cell diagram



A typical animal cell

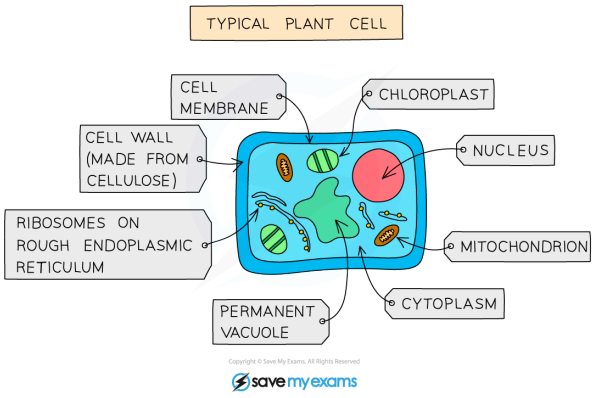
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- They contain a **nucleus** with a **distinct membrane**
- Cells have **cell walls** made out of **cellulose**
- They contain **chloroplasts** (so they can carry out **photosynthesis**)
- Carbohydrates are stored as **starch** or **sucrose**

Plant cell diagram

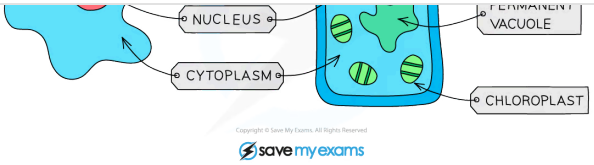


A typical plant cell

Plant and animal cell structure and function

Structure	Function
Nucleus	• Contains the DNA (genetic material) which controls the activities of the cell
Cytoplasm	• A gel like substance composed of water and **dissolved solutes** • Supports the internal structures of the cell • Site of many **chemical reactions** (including anaerobic respiration)
Cell membrane	• Holds the cell together separating the inside of the cell from the outside • Controls which substances **enter or leave** the cell
Ribosomes	• Found in the cytoplasm • The site of **protein synthesis**
Mitochondria	• The site of **aerobic respiration**

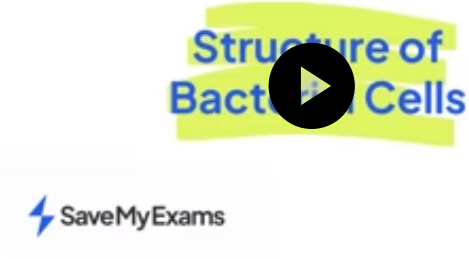
Cell structure diagram



An animal and plant cell as seen under a light microscope

Plant cell structure and function

Structure	Function
Cell wall	• Made of **cellulose** (a polymer of glucose) • Gives the cell extra **support**, defining its shape
Chloroplast	• Contains the green **chlorophyll** pigment that absorbs light energy for **photosynthesis**
Permanent vacuole	• Contains **cell sap**: a solution of sugar and salt • Used for storage of certain materials • Helps to **support** the shape of the cell



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Bacteria cells

Bacteria cell structure

- Bacteria, which have a wide variety of shapes and sizes, all share the following biological characteristics:
  - They are **microscopic single-celled organisms**
  - Possess a **cell wall** (made of **peptidoglycan**, not cellulose), **cell membrane**, **cytoplasm** and **ribosomes**
  - **Lack a nucleus** but contain a **circular chromosome of DNA that floats in the cytoplasm**

chromosomal DNA
    - They **lack mitochondria, chloroplasts and other membrane-bound organelles** found in animal and plant cells
- Some bacteria also have a **flagellum** (singular) or **several flagella** (plural). These are **long, thin, whip-like tails** attached to bacteria that allow them to **move**
- Examples of bacteria include:
    - **Lactobacillus** (a rod-shaped bacterium used in the production of yoghurt from milk)
    - **Pneumococcus** (a spherical bacterium that acts as the pathogen causing pneumonia)

Bacteria cell diagram

A typical bacterial cell

Identifying cell structures & function

- Within the cytoplasm, the following organelles are visible in almost all cells except prokaryotes when looking at higher magnification (ie using an electron microscope):
    - **Mitochondria** (singular: mitochondrion) are organelles found throughout the cytoplasm
    - **Ribosomes** are tiny structures that can be free within the cytoplasm or attached to a system of membranes within the cell known as **Endoplasmic Reticulum**
    - Endoplasmic reticulum studded with ribosomes looks **rough** under the microscope; this gives rise to its name of **Rough Endoplasmic Reticulum** (often shortened to **R.E.R.**)
    - **Vesicles** can also be seen using a higher magnification - these are small circular structures found moving throughout the cytoplasm

Identifying cell structures under a microscope

Structures in an animal cell visible under a light microscope and an electron microscope

Structures in a plant cell visible under a light microscope and an electron microscope

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